

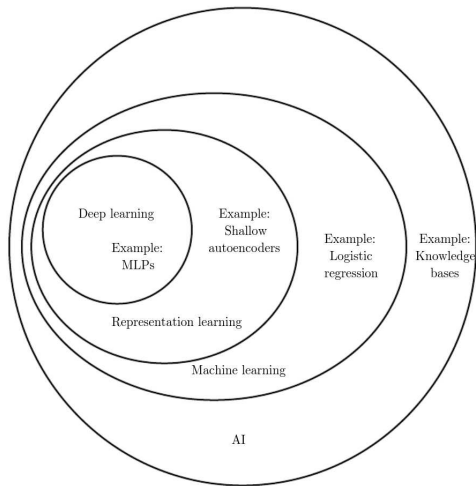
Brief History of Recent Deep Learning

AI for ecologists

Paul Tresson

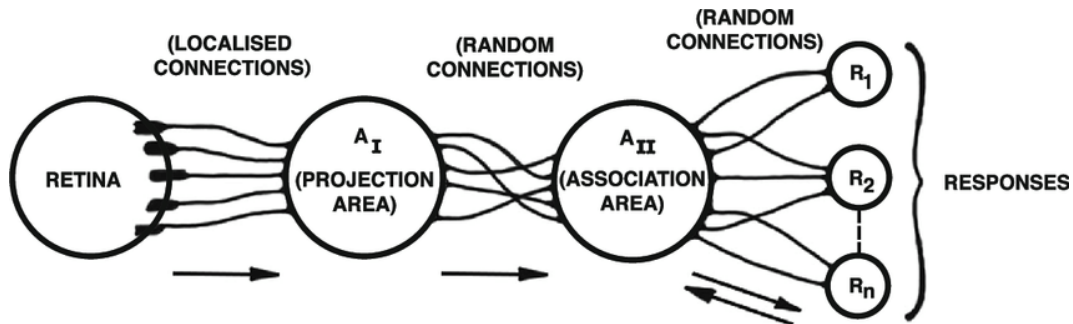
03/06/26

Reminder



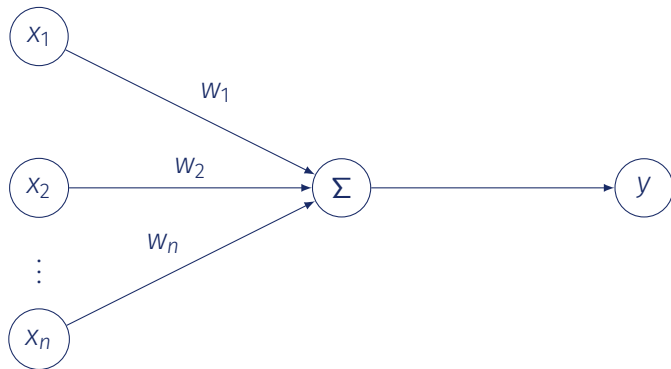
Different types of Artificial Intelligence. From Goodfellow et al., 2016

First Neural Network Model : The Perceptron



First probabilistic model of a neuron proposed by Rosenblatt, 1958

First Neural Network Model : The Perceptron



First Neural Network Model : The Perceptron

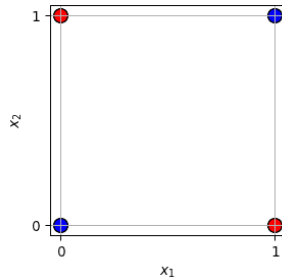
$$y = \begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

First Neural Network Model : The Perceptron

$$y = \begin{bmatrix} w_1 & w_2 & \dots & w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

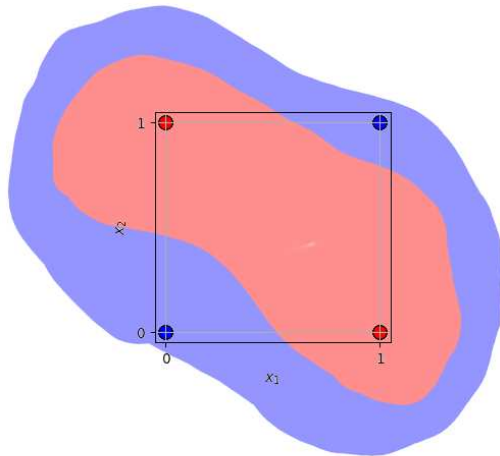
First Roadblock : XOR problem

x_1	x_2	y
0	0	0
1	0	1
0	1	1
1	1	0



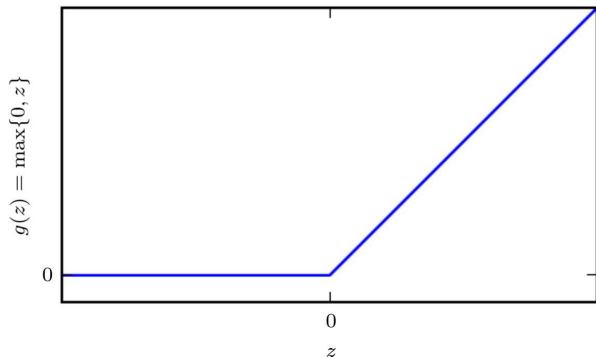
First Roadblock : XOR problem

x_1	x_2	y
0	0	0
1	0	1
0	1	1
1	1	0



1980's : First Deep Neural Networks

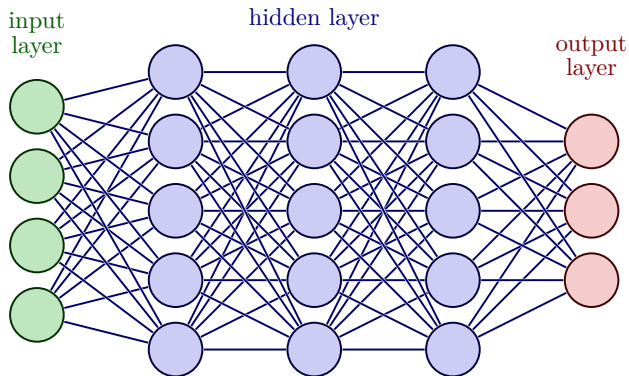
- Non-linearity (e.g. ReLu)



Rectified Linear Unit. Figure from Goodfellow et al., 2016

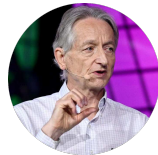
1980's : First Deep Neural Networks

- Non-linearity (e.g. ReLu)
- Multi Layer Perceptron

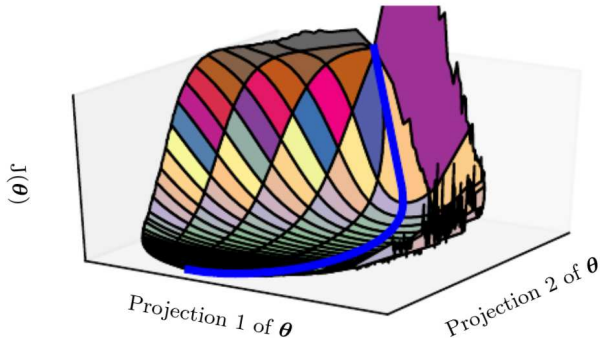


Multi Layer Perceptron

1980's : First Deep Neural Networks



- Non-linearity (e.g. ReLu)
 - Multi Layer Perceptron
 - Gradient Descent and Backpropagation
- Rumelhart, 1986



Gradient descent example. Figure from Goodfellow et al., 2016

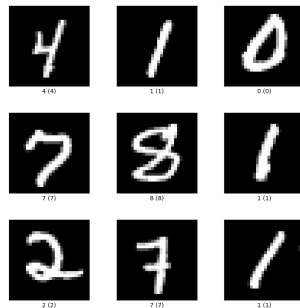
As matrices

$$y = \text{ReLU} \left(\begin{bmatrix} w_{1,1} & w_{1,2} & \dots & w_{1,n} \\ w_{2,1} & w_{2,2} & \dots & w_{2,n} \\ \vdots & & & \vdots \\ w_{k,1} & w_{k,2} & \dots & w_{k,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \right)$$

1990's : First Successes



- MNIST
LeCun, 1998

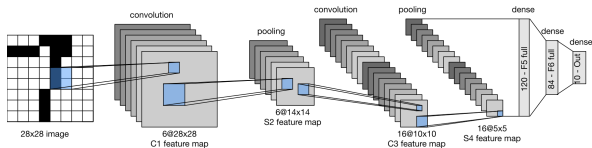


Images from MNIST

1990's : First Successes



- MNIST
LeCun, 1998
- LeNet
LeCun et al., 1989b



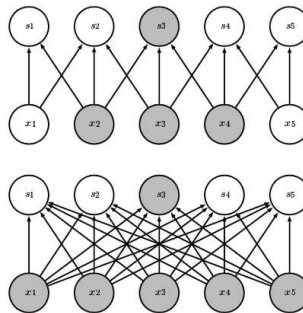
LeNet5.

Figure from Zhang et al. - <https://github.com/d2l-ai/d2l-en>

1990's : First Successes

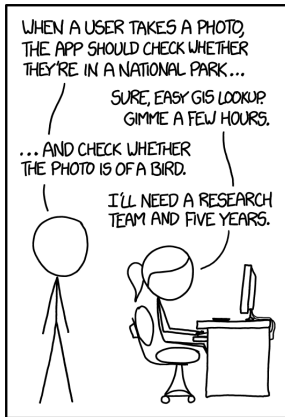


- MNIST
LeCun, 1998
- LeNet
LeCun et al., 1989b
- Convolutional Neural Network (CNN)
LeCun et al., 1989a



sparse connectivity with CNNs.
Figure from Goodfellow et al., 2016

Good proof of concept but too costly in computing power and datasets



IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

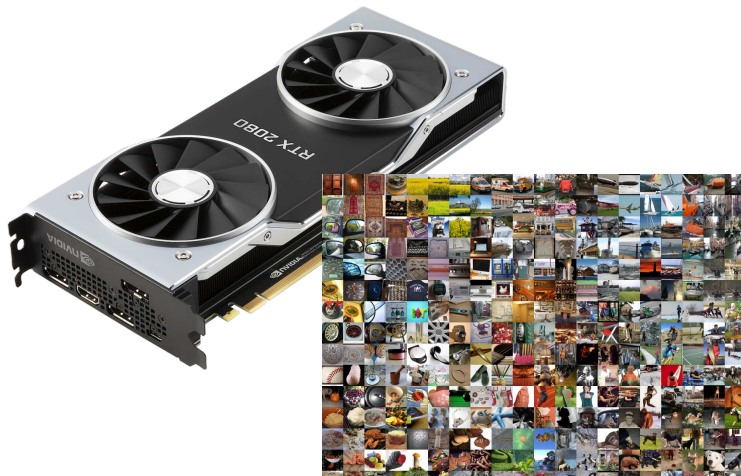
XKCD comic from 2014

Good proof of concept but too costly in computing power and datasets
...until

Good proof of concept but too costly in computing power and datasets
...until

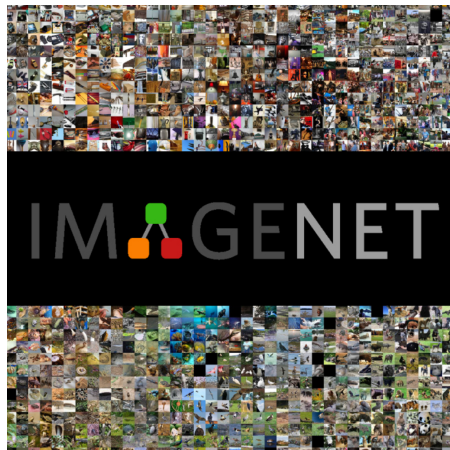


Good proof of concept but too costly in computing power and datasets
...until



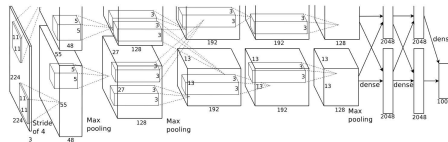
2010's : CNN revolution

- ImageNet
Deng et al., 2009



2010's : CNN revolution

- ImageNet
Deng et al., 2009
- AlexNet
Krizhevsky et al., 2012

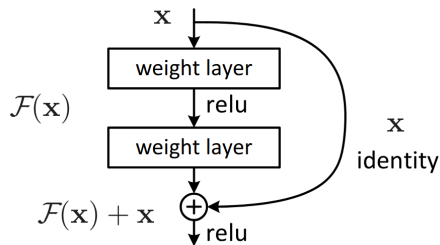


AlexNet. Figure from Krizhevsky et al., 2012

2010's : CNN revolution



- ImageNet
Deng et al., 2009
- AlexNet
Krizhevsky et al., 2012
- ResNet
He et al., 2016

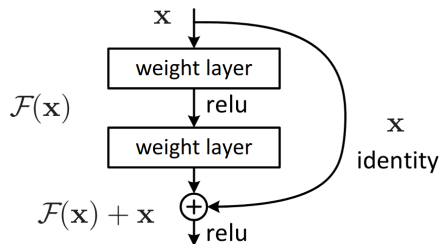


A residual connection. Figure from He et al., 2016

2010's : CNN revolution



- ImageNet
Deng et al., 2009
- AlexNet
Krizhevsky et al., 2012
- ResNet
He et al., 2016

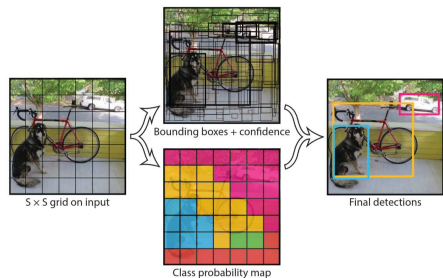


A residual connection. Figure from He et al., 2016

His 2016 paper *Deep Residual Learning for Image Recognition* is the most cited research paper in 5 years according to [Google Scholar](#)'s reports in 2020 and 2021.^{[7][8]}

2010's : CNN revolution

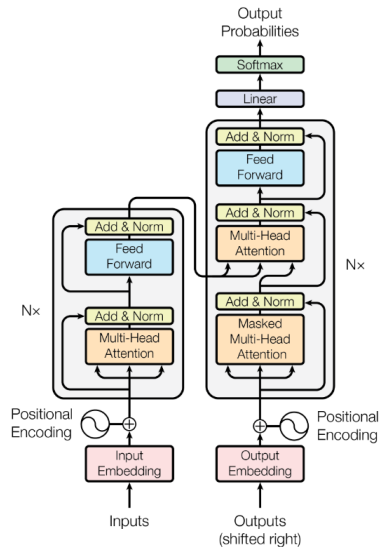
- ImageNet
Deng et al., 2009
- AlexNet
Krizhevsky et al., 2012
- ResNet
He et al., 2016
- YOLO, mask-RCNN...
He et al., 2017; Redmon et al., 2016



YOLO. Figure from Redmon et al., 2016

2020's : Transformers ...

- Attention is all you need
Vaswani et al., 2017



2020's : Transformers ... and Self Supervised Learning

- Attention is all you need
Vaswani et al., 2017
- LLMs (GPT, BERT...)
Devlin et al., 2019



2020's : Transformers ... and Self Supervised Learning

- Vision Transformer
Dosovitskiy et al., 2020

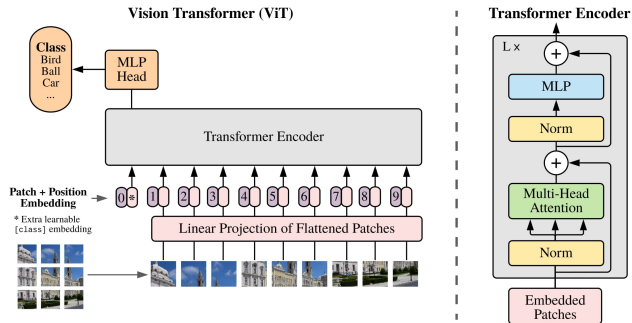


Figure from Dosovitskiy et al., 2020

2020's : Transformers ... and Self Supervised Learning



- Vision Transformer
Dosovitskiy et al., 2020
- Masked Auto Encoder
He et al., 2022

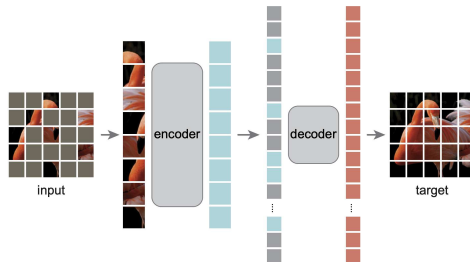


Figure from He et al., 2022

2020's : Transformers ... and Self Supervised Learning

- Vision Transformer
Dosovitskiy et al., 2020
- Masked Auto Encoder
He et al., 2022
- DINO
Caron et al., 2021



Figure from Caron et al., 2021

A lot left out !

Finetuning

A lot left out !

Finetuning

Games

A lot left out !

Reinforcement learning

Finetuning

Games

A lot left out !

Reinforcement learning

Finetuning

Games

Time series

A lot left out !

Reinforcement learning

Finetuning

Games

Time series

Speech recognition

A lot left out !

Reinforcement learning

Finetuning

Games

Image generation

Time series

Speech recognition

A lot left out !

Reinforcement learning

Finetuning

Games

Image generation

Diffusion models

Time series

Speech recognition

A lot left out !

Reinforcement learning

Finetuning

Games

Image generation

Diffusion models

Time series

Speech recognition

...

Useful resources

State of the art

- Huggingface
- PapersWithCode

Getting started

- Pytorch
- Keras

Understanding papers

- Yannic Kilcher
- AI coffe break

Understanding visually

- 3blue1brown
- deepia

Thanks for you attention !

Let's practice !

References i

- Caron, Mathilde et al. (2021). **“Emerging properties in self-supervised vision transformers”**. In: *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9650–9660.
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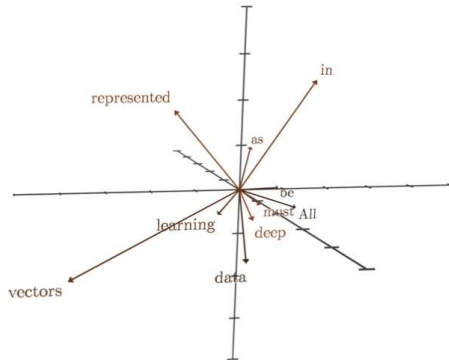
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- Redmon, Joseph, Santosh Divvala, Ross Girshick, and Ali Farhadi (2016). **“You only look once: Unified, real-time object detection”**. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 779–788.

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- Rosenblatt, Frank (1958). **“The perceptron: a probabilistic model for information storage and organization in the brain.”**. In: *Psychological review* 65.6, p. 386.
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- Vaswani, Ashish et al. (2017). **“Attention is all you need”**. In: *Advances in neural information processing systems* 30.

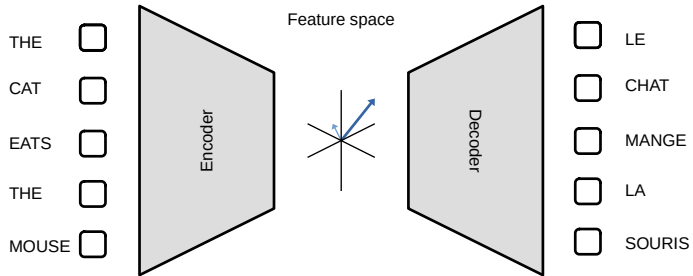
Sidestep : Transformers

All
data
in
deep
learning
must
be
represented
as
vectors

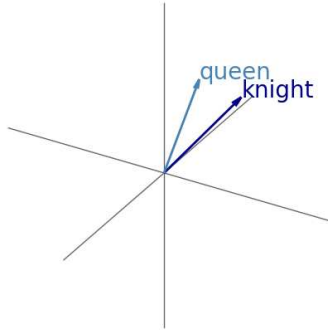


Adapted from 3blue1brown

Sidestep : Transformers

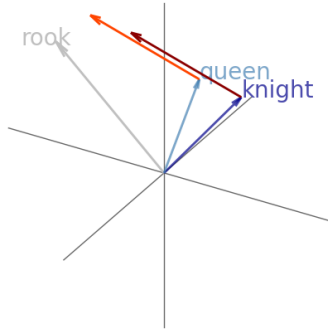


Sidestep : Transformers



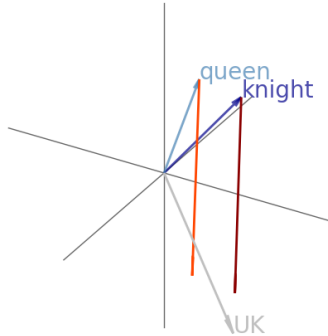
The knight saves the queen

Sidestep : Transformers



Queen takes knight , **checkmate !**

Sidestep : Transformers



He was knighted by Queen Elizabeth II

Sidestep : Transformers

